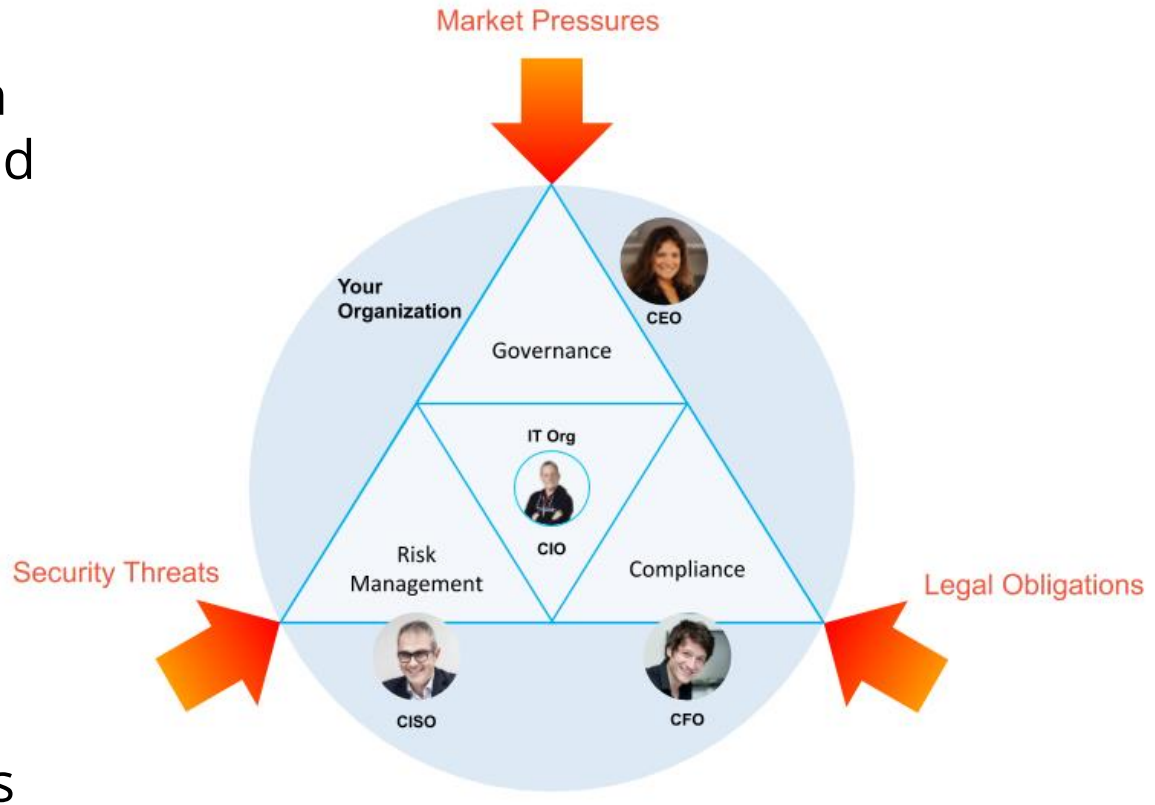
DevSecOps technology governance

DevSecOps technology governance

By addressing these governance topics with clear objectives, strategies, and tools, organizations can effectively manage DevSecOps pipelines, tools, and infrastructure while ensuring compliance and operational efficiency.

- DevSecOps pipeline sprawl
- DevSecOps tool sprawl
- DevSecOps infrastructure cost governance
- DevSecOps governance auditing
- integrating DevSecOps governance into metrics and dashboards.



Source: https://thenewstack.io/devsecops-can-address-the-challenges-of-governance-risk-compliance-grc/

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DevSecOps pipeline sprawl

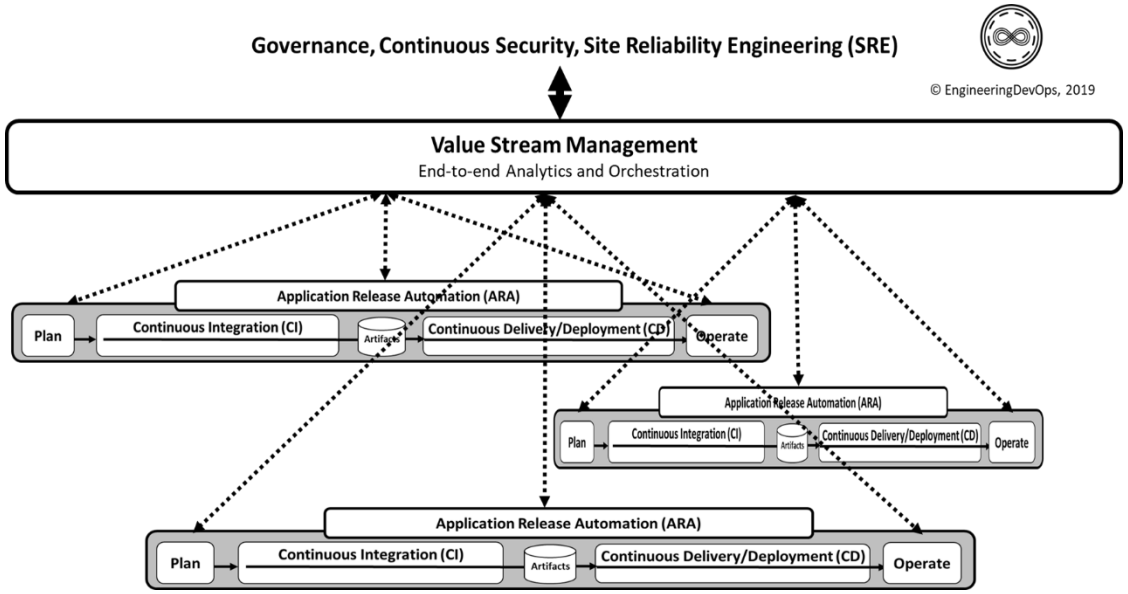
To manage and standardize the increasing number of CI/CD pipelines across teams and projects, ensuring security and consistency without overwhelming resources.

Challenges:

- **Fragmentation:** different teams may create pipelines using inconsistent tools and configurations, leading to inefficiencies and gaps in security.
- **Complexity:** managing dependencies and integrations across pipelines becomes harder as they scale.

Solutions:

- **Pipeline standardization:** use templates and modular pipeline designs to enforce best practices.
- **Governance policies:** implement clear rules for pipeline creation, usage, and decommissioning.
- **Centralized orchestration:** Tools like Plutora, Jenkins, GitLab CI/CD to centralize management.



Source: www.engineeringdevops.com

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DevSecOps tools sprawl

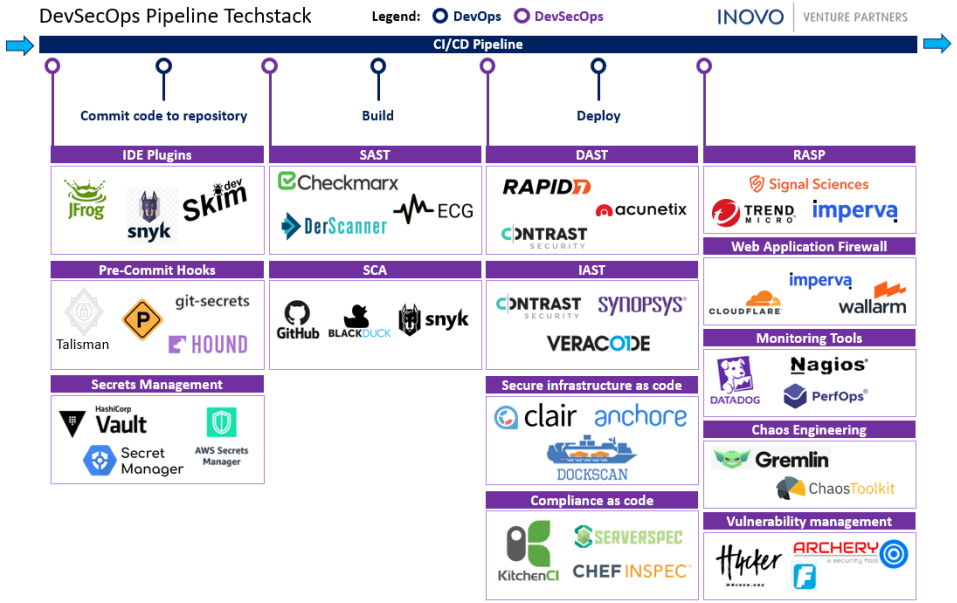
To prevent the proliferation of redundant or unintegrated tools, which can complicate workflows, increase costs, and create security risks.

Challenges:

- **Overlapping tools:** teams may adopt multiple tools with similar functionalities, leading to inefficiency.
- **Lack of integration:** disconnected tools hinder collaboration and data sharing.
- **Training burden:** more tools require additional training, increasing the cognitive load for teams.

Solutions:

- **Tool rationalization:** conduct periodic audits to identify redundant tools and consolidate into a core toolset.
- **Tool governance board:** evaluate and approve new tools based on organizational needs.
- **Integration standards:** ensure tools communicate effectively.



Source: https://blog.mergify.com/automating-security-for-applications-with-devsecops/

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DevSecOps infrastructure cost governance

To monitor and control infrastructure costs while maintaining performance and security in DevSecOps environments.

Challenges:

- **Cloud cost overruns:** excessive resource usage and improper scaling can inflate costs.
- **Untracked resources:** orphaned or underutilized resources often go unnoticed.
- **Balancing cost and performance:** over-optimizing for cost may degrade performance or security.

Solutions:

- **Cost monitoring tools:** track usage and identify inefficiencies.
- **IaC policies:** implement tagging and auto-scaling policies in IaC templates.
- **Regular reviews:** conduct cost reviews to align spending with business priorities.



Source: https://www.jamcracker.com/blogs/cloud-cost-management-strategies

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DevSecOps governance auditing

To ensure compliance with organizational, regulatory, and security standards through regular assessments of processes and configurations.

Challenges:

- **Dynamic environments:** frequent changes in pipelines and configurations make auditing complex.
- **Manual processes:** traditional audits can be time-consuming and prone to human error.
- **Compliance fatigue:** teams may view audits as a burden, leading to resistance.

Solutions:

- **Automated audits:** continuously monitor and generate compliance reports.
- **Pre-built audit frameworks:** enforce and audit policies automatically.
- **Auditing dashboards:** build governance dashboards to visualize audit results in real-time.



Source: https://bobcares.com/blog/devsecops-security-checklist/

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DevSecOps governance dashboards

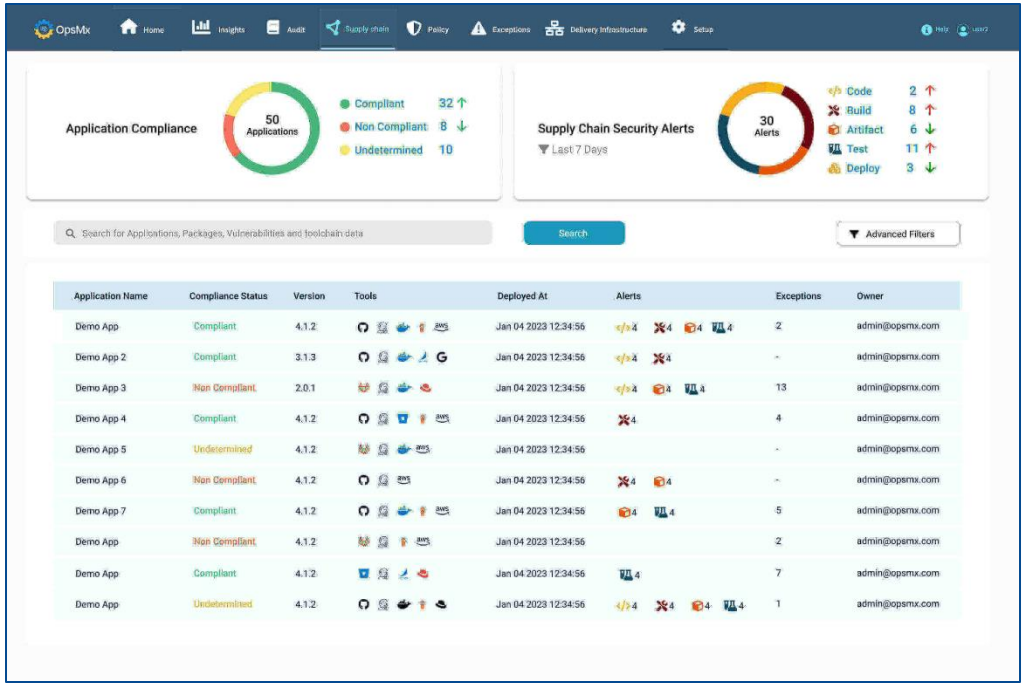
To provide visibility into governance-related metrics, enabling data-driven decisions and continuous improvement.

Challenges:

- **Metric overload:** too many metrics can overwhelm teams, diluting focus on critical governance issues.
- **Data silos:** governance metrics may reside in disconnected tools, making consolidation difficult.
- **Real-time insights:** ensuring dashboards are updated in real time for effective decision-making.

Solutions:

- **Balanced metrics:** focus on actionable governance metrics like pipeline security success rates, tool utilization, and infrastructure cost trends.
- **Integrated dashboards:** use tools like Grafana or Power BI to aggregate governance data across the toolchain.
- **Automated updates:** leverage CI/CD hooks and APIs to feed governance metrics into dashboards dynamically.



Source: https://www.opsmx.com/secure-continuous-delivery/security-and-governance-for-secure-cd/

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